

Zhengyao Huang

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D.O.B.: June 24, 2001 in Hefei PRC

Personal Profile

I am a first-year PhD student in Mathematics at Durham University. My supervisors are Wilhelm Klingenberg and Fernando Galaz-García.

Research Interests

My research interests lie in the Calculus of Variations and in Geometric Measure Theory. I currently work on the Optimal Transport problem and on decomposition of vector charges.

Education

2019-2022 BSc in Physics

University of Wisconsin, Madison

GPA: 3.83/4.00

2022 – 2024: MSc in Theoretical Physics

Utrecht University, Utrecht

GPA: 7.80/10.0

Thesis: Non-Equilibrium Dynamics of the Tomonaga-Luttinger Liquid and Exceptional Points of Parafermion.

2024 – 2025: MSc in Pure Mathematics

Durham University, Durham

GPA: First-Class Honours

Thesis: Solving the Monge Problem by Disintegration of Transport Sets.

2025 – present: PhD in Mathematics

Durham University, Durham

Supervisor : Wilhelm Klingenberg

Publications

A Sudakov Decomposition in Riemannian Manifolds with Positive Curvature, 2026 [arXiv:2603.18935]

Zhengyao Huang, submitted to Mathematische Nachrichten

Smirnov Decomposition of a Horizontal Vector Charge in the Heisenberg Group, 2026 [to be posted on arXiv] *Zhengyao Huang, Wilhelm Klingenberg*

Skills

Python, Mathematica

Languages: English (Fluent), Chinese (Native)

References

Fernando Galaz-Garcia, Durham University, fernando.galaz-garcia@durham.ac.uk

Wilhelm Klingenberg, Durham University, wilhelm.klingenberg@dur.ac.uk